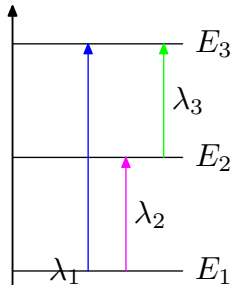


Absorption

Emission

 $E$  $E_3$  $E_2$  $E_1$  $\lambda_3$  $\lambda_2$  $\lambda_1$ 

$$\Delta E = E_3 - E_1$$

$$E_{\text{Photon}} = h \cdot f = \frac{h \cdot c}{\lambda_1}$$

 $E_3$  $E_2$  $E_1$  $\lambda'_2$  $\lambda'_1$  $\lambda'_3$ 

$$\Delta E = E_3 - E_1$$

$$E_{\text{Photon}} = h \cdot f = \frac{h \cdot c}{\lambda'_1}$$

